other functions of n such that $T_1(n) \leq f(n) < O(n^2)$ for n large enough.

Likewise, we could also say that $T_1(n) = O(n^{1000})$: that would certainly be true, but how helpful would it be knowing that our algorithm takes less than the age of the Universe on a small input[1]?

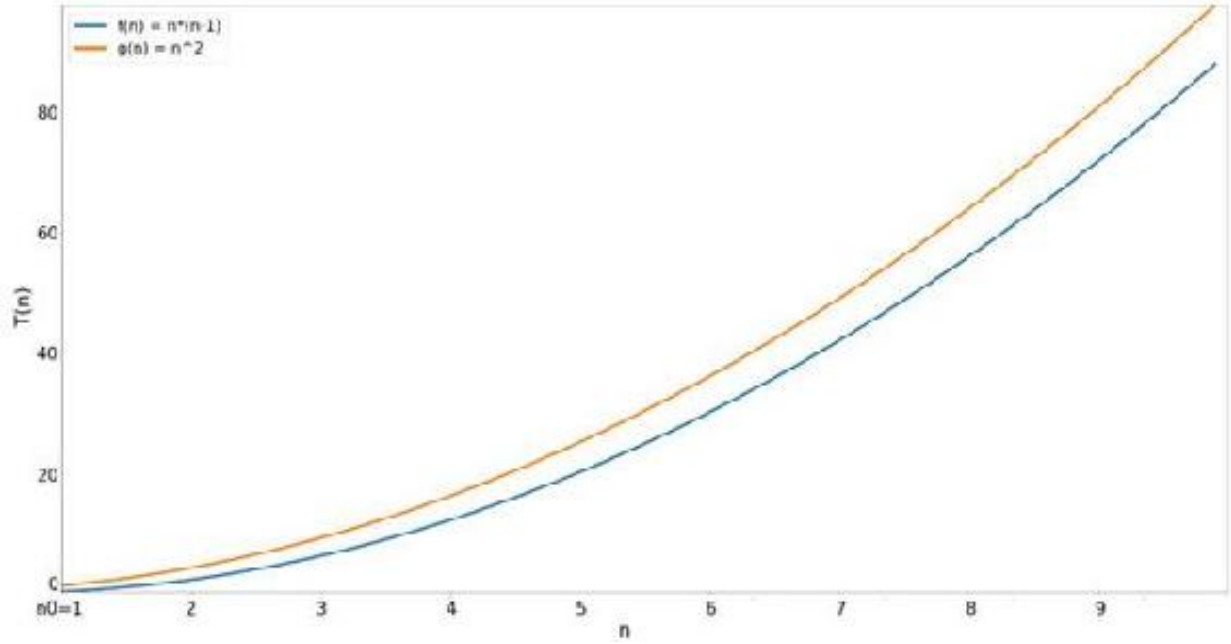



**Figure B.4 Visual comparison of functions: `f(n)=n*(n-1)`, `g(n) = n^2`. The latter is always large for any positive value of `n`.**

As you might suspect, we are usually interested in strict bounds. While this is true and expected, and for our summation examples it seems trivial proving that the `O(n)` bound is strict, you might be surprised to learn that there are some algorithms for which we don't know what the stricter bound is, or at least we can't prove it.